Manuel Solalinde

Senior Full Stack Engineer - AI/LLM Applications - MSc. Applied Mathematics

Email / CV / LinkedIn / Github1 / Github2 / Stackoverflow / Website

🏢 Work Experience

Senior Software Engineer - BairesDev - May 2022 - present

- Full stack engineer on TriformaIQ Real Estate Studio, an AI-powered multi-tenant SaaS platform for commercial real estate. It combines GIS-based site selection with weighted multi-criteria scoring, entitlements, due diligence, document management and audit-ready records.
- Built LLM-powered document analysis features using retrieval-augmented generation (RAG): document ingestion and chunking, vector embeddings and semantic search, structured LLM outputs grounded in source citations, and regression evaluation of prompt quality.
- Implemented subscription billing and licensing: payment provider integration via webhooks, subscription lifecycle management, and plan-based feature gating.
- Developed role-based approval workflows, per-user and per-organization rate limiting of third-party APIs, and geospatial data features with tenant data isolation.
- Designed AI-assisted development workflows with Claude Code (custom skills and multi-agent orchestration) that automate ticket analysis, implementation, code review and end-to-end verification.
- Involved in development of web applications and data science projects at BKV Corporation, focusing on creating intuitive surveillance dashboards and engineering tools.
- My role includes analyzing and interpreting extensive datasets, employing advanced methods to extract actionable insights
- Technologies: Python, FastAPI, SQLModel, PostgreSQL/PostGIS, pgvector, Alembic, Celery, Redis, Docker, Anthropic API, Stripe, ESRI ArcGIS, Auth0, React, TypeScript, Pytest, Playwright.

Senior Software Engineer - BKV Corporation - May 2022 - June 2026

- Designed and developed machine learning applications for optimization of artificial lift systems.
- Developed AI-powered applications leveraging large language models for event detection and data structuring in gas well operations.
- Built web applications for real-time visualization, analysis, and processing of gas wells data, to facilitate insightful data interpretation and decision-making for engineers and managers.
- Significantly enhanced the efficiency of existing data processing algorithms, achieving over 10x improvement in performance. This set of optimizations led to significant increase in the usage of related applications.
- Developed templates and libraries, streamlining the process of creating new applications and reusing existing components.
- Architected and implemented AWS Infrastructure Solutions. Development of robust libraries and templates for seamless deployment of infrastructure into AWS, leveraging advanced infrastructure as code techniques with pulumi.
- Technologies: Python, Javascript, Fastapi, React, Azure AD, Plotly, Redux, Postgres SQL, AWS Cloud Computing Services, Nodejs, HTML/CSS. Testing: Cypress, Jest (frontend), Pytest (backend).
- Data Science libraries: scikit-learn, pandas, numpy, plotly, tensorflow, keras, streamlit.

Software Developer - Self Employed / Independent Contractor - Jul 2017 - Jul 2022

- Full Stack Developer for Centralhub Search Engine Platform. A search engine platform based on web scraping automation of regional e-commerce sites.
- Developer of a tool for converting dual fisheye camera videos to equirectangular stitched videos using NVIDIA CUDA, NVDEC, NVENC.
- Developer of a management system using Excel VBA and SQL.
- Backend developer of ecommerce sites.
- Worked in a cross-functional, agile team to rebuild the SPR system for the Paraguayan National Secretary of Planning.
- Assistance with information architecture and UX writing for webapps.

Software Developer - Engineer - IMPSA - Mendoza, Argentina - Jan 2011 - Jul 2013

- Lead Developer of the ARTUR platform (C++ Language), used for the calculation and pre-dimensioning of hydraulic turbine structural component elements.
- Assisted in the development of design manuals that set the criteria for calculation and dimensioning of water turbine structural elements.

💻 Projects

TriformaIQ Real Estate Studio - project details
An AI-powered multi-tenant SaaS platform for commercial real estate that unifies site selection, entitlements, due diligence and audit-ready documentation.

- *Client*: TriformaIQ (via BairesDev)
- *Focus*: Full Stack, AI/LLM Features, SaaS Infrastructure
- *Key Features*:
  - LLM-powered document analysis with RAG: ingestion and chunking, vector embeddings and semantic search, citation-grounded structured outputs, and regression evaluation of prompt quality.
  - Subscription billing and licensing with webhook-driven lifecycle management and plan-based feature gating.
  - Role-based approval workflows, per-user and per-organization API rate limiting, and geospatial scoring features with strict tenant isolation.
  - AI-assisted engineering workflows with Claude Code (custom skills and multi-agent orchestration).
- *Technologies*: Python, FastAPI, PostgreSQL/PostGIS, pgvector, Celery, Redis, Docker, LLMs, Stripe, ESRI ArcGIS, Auth0, React, TypeScript, Pytest, Playwright.

AI Agent Chat - project details

A full-stack conversational AI agent that answers oil & gas operational questions over the corporate well database and internal documentation, built on Amazon Bedrock AgentCore.

- *Client*: BKV Corporation
- *Focus*: Full Stack, AI Agents, Cloud Infrastructure
- *Key Features*:
  - Natural-language to read-only SQL over the corporate well-data warehouse, with cited, tool-backed answers.
  - Real-time streaming (SSE) of tool calls, inputs, and results; persistent conversation history with auto-generated titles and feedback.
  - Gateway (MCP) tool suite: SQL + schema docs, internal datadocs hybrid search, web search/fetch, and a Python code interpreter.
  - Four interchangeable agent patterns (Strands, LangGraph, Claude Agent SDK single & multi-agent) on Bedrock AgentCore Runtime (Claude Sonnet 4.5).
  - React 19 SPA + FastAPI on ECS Fargate; Microsoft Entra ID (OIDC) user auth and Cognito M2M; provisioned as code with Pulumi.
- *Technologies*: Python, FastAPI, TypeScript, React, Tailwind CSS, Amazon Bedrock AgentCore (Runtime, Gateway, Memory, Code Interpreter), Strands Agents, LangGraph, Claude Agent SDK, AWS (ECS Fargate, ALB, Lambda, DynamoDB, Cognito), Pulumi, Docker, Microsoft Entra ID.

Gas Injection Optimization App - project details
A data-driven application that recommends optimal gas-injection rates for gas-lift wells and ranks wells by how far each has drifted from its recommendation.

- *Client*: BKV Corporation
- *Focus*: Data Science
- *Key Features*:
  - Recommendation model: history clustering (DecisionTreeRegressor) plus a per-cluster penalty score.
  - Data-driven penalty weights fitted against engineer-labeled wells via Bayesian optimization.
  - Report ranking wells by injection-vs-recommendation deviation, with operator overrides.
  - Parallel Lambda fan-out for on-demand report regeneration; per-well data in S3 (parquet).
- *Impact*: Reduced over-injection on flagged wells, contributing to a dramatic improvement in net sales.
- *Technologies*: Python, Streamlit, Pandas, NumPy, Scikit-learn, scikit-optimize, Plotly, AWS (S3, Lambda, DynamoDB), Pulumi, Parquet.

Surveillance Dashboard Application - project details
A set of web applications to visualize, analyze and process real time data from gas wells. Drastically improves performance over custom Power BI applications that serve similar purposes. Comprised of 3 sub applications: Single well, Multi well, Daily and Engineering Plots.

- *Client*: BKV Corporation
- *Focus*: Full Stack, Cloud Infrastructure, CI/CD
- *Key Features*:
  - Azure AD authentication.
  - Support user comments on time series data.
  - Automatic data resampling for time series data.
  - Selection and filter of wells from lists or maps.
  - Multi well selection and data comparison.
  - Can add smoothing and indicators to time series data.
- *Technologies*: Python, Javascript, Fastapi, React, Azure AD (Corporate authentication), Plotly, Redux, Postgres SQL, AWS Cloud Computing Services, HTML/CSS. Testing: Cypress, Jest (frontend), Pytest (backend).

Gas Targets App - project details
A full-stack web application that generates and sets the monthly gas-production targets for every well, combining a purpose-built trend tool with guided manual review of edge cases.

- *Client*: BKV Corporation
- *Focus*: Full Stack, Cloud Infrastructure
- *Key Features*:
  - Four-stage monthly workflow: analysis selection, DB update, trend generation, override review, and final CSV export.
  - Parallel, queue-driven Lambda fan-out (SQS) processing the full well set per analysis.
  - Reads raw production history from a cross-account Amazon Redshift warehouse into a private PostgreSQL working DB.
  - Case-study reviews (last-7-days production, trend review) with Excel-based engineer overrides.
  - Cloud-native stack provisioned as code (Pulumi); Dockerized services deployed to App Runner / Lambda via GitHub Actions and ECR.
- *Technologies*: Python, JavaScript, React, Redux, FastAPI, PostgreSQL, Amazon Redshift, AWS (App Runner, Lambda, SQS, RDS, VPC, ECR), Pulumi, Docker, GitHub Actions, Azure AD, Plotly.js.

Plunger Optimization App - project details
A data analytics platform to optimize and diagnose plunger-lift gas wells from real-time SCADA data, combining signal processing, ML classification and interactive visualization.

- *Client*: BKV Corporation
- *Focus*: Data Science, Full Stack
- *Key Features*:
  - Interactive plunger-cycle analysis and stacked-cycle visualization (Streamlit + Plotly).
  - Automated diagnostic report suite (load-factor, pressure sustained-increase, data quality).
  - ML-based continuous-vs-conventional well classification.
  - Daily fleet-wide report generation on AWS (ECS Fargate, Lambda, S3, CloudWatch).
- *Technologies*: Python, Streamlit, Pandas, NumPy, SciPy, Scikit-learn, Plotly, AWS (S3, ECS Fargate, Lambda, ECR), Pulumi, Docker, GitHub Actions.

BKV Pulumi Templates - project details

An internal infrastructure-as-code library and project scaffolding that standardizes how the team deploys containerized applications to AWS ECS Fargate — the deployment foundation behind the client's applications.

- *Client*: BKV Corporation
- *Focus*: Cloud Infrastructure
- *Key Features*:
  - Reusable `ECSApplicationStack` Pulumi component: ECR, ALB, Route53, ACM, WAF, IAM, CloudWatch, Fargate service, and autoscaling in one stack.
  - Corporate SSO built in — Azure AD (OIDC) at the load balancer and Azure AD SAML SSO federation for Athena access.
  - Composable add-ons for Athena and cross-account Redshift access.
  - `bkv-ecs-aws-python` project template with config scripts and GitHub Actions CI/CD (preview-on-PR, deploy-on-merge).
- *Technologies*: Python, Pulumi, AWS (ECS Fargate, ALB, ECR, Route53, ACM, WAF, CloudWatch, IAM, S3, Athena, Glue, Redshift, Secrets Manager), Azure AD (OIDC + SAML SSO), Docker, GitHub Actions.

BKV Streamlit Apps - project details

A multi-page Streamlit platform bundling a growing collection of reservoir and production engineering tools for gas wells, sharing one S3 parquet data pipeline and a single containerized ECS Fargate deployment.

- *Client*: BKV Corporation
- *Focus*: Data Science, Cloud Infrastructure
- *Applications*:
  - Resolution Visualizer — visualizes SCADA sensor reading frequency per well, by day and channel.
  - Salt Water Injection — infinite-reservoir radial pressure-propagation calculator (Theis equation).
  - Well Communication — detects inter-well pressure interference from correlated production residuals.
  - Critical Rate Sensitivities — computes the minimum gas rate required to lift liquids out of the wellbore.
  - CO2 Runtime History — real-time sensor dashboard for the active CO2 injection sites.
  - Horner Plots — pressure-transient analysis on shut-in data, estimating reservoir permeability.
  - Hall Plots — injection-well performance from cumulative pressure vs. cumulative injected volume.
- *Technologies*: Python, Streamlit, Plotly, Pandas, NumPy, SciPy, Scikit-learn, AWS (ECS Fargate, Lambda, S3, DynamoDB, ECR, CloudWatch, Bedrock), Pulumi, Docker, GitHub Actions, Parquet, MkDocs.

Centralhub Search Engine - project details
A search engine platform based on web scraping of local e-commerce sites.

- *Focus*: Full Stack, Web Scraping
- *Description*: A search engine platform based on web scraping automation of regional e-commerce sites. Consists of 5 sub-projects: 1. An E-commerce Search Engine Frontend, 2. E-commerce Search Engine Admin Backend, 3. Web scraping API core and tools, 4. Web scraping automation and synchronization tools, 5. Devops, Search Engine Implementation, Architecture, and Integrations.
- *Technologies*: Elasticsearch, Python, Flask, Docker, NoSQL, Scrapy, Selenium, Firebase (Functions, Firestore, Authentication, Hosting), Git, JavaScript, React, Redux, Nodejs, Google Cloud Platform (PubSub, Task Scheduler, Compute Engine, Cloud Run, Task Runner, Logging). Testing: Jest, Pytest.

Dual Fisheye to VR180 - Nvidia GPU Video Stitcher - project details
A tool for converting dual fisheye camera videos to equirectangular stitched videos using NVIDIA CUDA, NVDEC, NVENC.

- *Focus*: GPU Computing
- Results obtained were x100+ faster than those obtained using serial CPU programming.
- *Technologies*: C++, Cmake, CUDA, OpenGL, OpenCV, Nvidia VRWorks.

Artur Platform - Predimensioning of hydraulic turbine structural components - project details

A tool for predimensioning of hydraulic turbine structural components.

- Lead developer for the ARTUR platform, used for the calculation and pre-dimensioning of hydraulic turbine structural component elements.
- Assisted in the development of design manuals that set the criteria for calculation and dimensioning of water turbine structural elements.
- *Employer*: IMPSA - Juan De Miguel
- *Focus*: Scientific Computing
- *Technologies*: C++, ANSYS Multiphysics API, Matlab, Excel VBA, SolidEdge API.

Enertrag WindEnergy-Hydrogen-Biogas Integrated System Simulation Tool - project details

A tool for simulation of wind energy systems using hydrogen and biogas.

- Developed in coordination with the Hochschule Stralsund as a base tool for simulations in the field of Wind Power - Hydrogen integrated systems for the ENERTRAG company.
- *Employer*: Hochschule Stralsund - Prof. Dr. rer. nat. Jochen Lehmann - Prof. Dr.-Ing. Thomas Luschtinetz
- *Focus*: Scientific Computing, Optimization
- *Technologies*: Matlab, Matlab Genetic Algorithms Toolbox.

SPR System - Technical Secretary of Planning - Paraguayan Government - project details

Design specifications and UI/UX for the SPR system - Technical Secretary of Planning - Paraguayan Government (STP).

- Worked in a cross-functional, agile team to rebuild the SPR system
- Assistance with information architecture and UX writing for the web app
- *Client*: STP-Paraguay
- *Focus*: UX/UI, Information Architecture

StuffILearned - Personal Software development Notes - project details

A software development personal blog to keep track of new technologies, courses and personal notes.

- Provide easy access to common software development notes.
- *Focus*: Web Development
- *Technologies*: PHP, Wordpress, MySQL, html, css.

Matchmaking - project details

A simple flask web application to balance team players for video games.

- Small open source project.
- *Github*: https://github.com/manolosolalinde/matchmaking
- *Focus*: Web Development
- *Technologies*: Python, Flask, MySQL, SQLAlchemy, Jinja, Bootstrap, html, css.

A Finite Element Based Level Set Approach for the Simulation of Two-Phased Flows - short summary

A Finite Element Based Level Set Approach for the Simulation of Two-Phased Flows.

- *Description*: **Master Thesis**. Level set approach for the simulation of two-phase immiscible newtonian fluids.
- *Github*: https://github.com/manolosolalinde/FEM-Multiphase
- *Dissertation*: https://www.bdtd.uerj.br:8443/bitstream/1/11742/1/Nestor%20M%20Solalinde.pdf
- *Focus*: Scientific Computing, Numerical Simulation
- *Technologies*: C++, Matlab, Wolfram Mathematica, Libmesh Finite Elements Library.

👩‍🎓 EDUCATION

MSc. in Mechanical Engineering – Applied Mathematics
UERJ - Rio de Janeiro, Brazil (2009-2011)

- Finished with straight A's.
- *Specialization*: Scientific computing, continuum mechanics, finite elements, optimization, fluid mechanics, heat transfer.
- *Master Thesis*: A finite element level-set approach for the simulation of two-phased flows. https://www.bdtd.uerj.br:8443/bitstream/1/11742/1/Nestor%20M%20Solalinde.pdf

Electromechanical Engineer
Universidad Nacional de Asunción - Asunción, Paraguay (2001-2008)

- Finished as a member of the honor roll (top 5 students).
- Initial enrollment test: rank 3 of 120 students admitted. (800 applicants).
- Auxiliary Teacher of Vector Mechanics (third semester subject).
- Course concluded with an average grade of 4.23 in a scale from 1 to 5, 5 being the highest grade.
- *Thesis*: Analysis of Electrolytic Hydrogen Production in Paraguay.

International Exchange Semester
Hochschule Stralsund - Stralsund, Germany (2007)

- Finished a 6 months course on Renewable Energies and Hydrogen Technology, top of the class, with an average of 1.27 in a scale from 1 to 5, 1 being the highest grade.

High School
San Ignacio de Loyola School - Asuncion, Paraguay (1995-2000)

- Private High School.
- General average grade in secondary education: 4.26 on a scale from 1 to 5, 5 being the highest grade.

💬 Languages

English: Fluent
Portuguese: Advanced
German: Intermediate
Spanish: Native

📚 Informal Software Engineering Education

Build a Backend REST API with Python & Django - Advanced

- Udemy - Course details
- Course duration: 8hs. Online Course.

Complete Guide to ElasticSearch

- Udemy - Course details
- Course duration: 12hs. Online Course.

Intro to Parallel Programming

- Coursera – Udacity - Course details
- Course duration: 21hs. Online Course.

Machine Learning

- Coursera – Standford University - Course details
- Course duration: 55hs. Online Course.

CS50's Web Programming with Python and JavaScript

- edX – Harvard University - Course details
- Course duration: 90hs. Online Course.

HTML, CSS, and Javascript for Web Developers

- Coursera – Johns Hopkins University - Course details
- Course duration: 33hs. Online Course.

Computer Vision Intro™ OpenCV4 in Python with Deep Learning

- Udemy - Course details
- Course duration: 10hs. Online Course.

Convolutional Neural Networks for Visual Recognition

- Standford University - Course details
- Course duration: 34hs. Online Course.

Complete Guide to TensorFlow for Deep Learning with Python

- Udemy - Course details
- Course duration: 14hs. Online Course.

OCW 6001 Introduction to Computer Science and Programming in Python

- MIT OCW - Course details
- Course duration: 12hs. Online Course.

C++

- TheChernoProject/Youtube - Course details
- Duration: 15hs. Online Course.

Introduction Courses

Most of these where used as technology introduction material before diving into deeper documentation.
GraphQL - The Net Ninja - Course details
Gatsby Tutorial – The Net Ninja - Youtube - Course details
Next.JS - The Net Ninja - Youtube - Course details
React Native - The Net Ninja - Course details
Javascript Unit Testing with Jest - Traversy Media - Course details
React, Redux & Firebase - The Net Ninja - Course details
Python REST API - Tech With Tim - Course details
Asynchronous Python - Live Python - Course details
React Redux - The Net Ninja - Course details
Firebase Firestore - The Net Ninja - Course details
Docker - freeCodeCamp.org - Course details
Docker Compose - Jake Wright - Course details

🦾 Electromechanical Engineering experience

Self-Employed / Consultant
January 2017 to current date

- Asuncion – Paraguay (January 2017 to current date)
- Calculation and assessment of structural components using finite element analysis tools for:
  - Naval industry (tugboats, barges).
  - Cement Industry (mills, structural components).
  - Shipyards.
- Design and calculation of hydraulic systems (industrial plants).
- Design and calculation of general steel structures.
- Electrical, HVAC and Plumbing projects.

Engineer / Manager @ SCE S.A. Asuncion – Paraguay (2016-2017)

- Engineering and Design: Design and calculation of steel structures.
- Electrical, HVAC and Plumbing projects for houses, several story buildings and residential complex.
- Management: Construction of a residential complex of 1800m2. Development of excel/SQL tools for management.

Engineer @ TECNOEDIL S.A.
Asuncion – Paraguay (2013-2015)

- BIM Management and Implementation with Revit Architecture and Revit MEP for the Engineering technical office.
- Coordination of all the task related electrical systems, HVAC, Fire protection and Plumbing, related to the construction of the Regional Hospital of Ciudad del Este. Client: IPS.

Engineer/Developer @ IMPSA
Mendoza – Argentina (2011 to 2013)

- Lead Developer of the ARTUR platform (C++ Language), used for the calculation and pre-dimensioning of hydraulic turbine structural component elements.
- Assisted in the development of design manuals that set the criteria for calculation and dimensioning of water turbine structural elements.

Engineering Intern @ TERMECO S.R.L.
Asuncion- Paraguay (Apr/2007-Jul/2007)

- Intern. Mechanical Designer, drawer.
- *Tasks*: Industrial Boilers design using Solidworks, Bioethanol Plant Project related design tasks.

Mechanical Designer @ TEDEC S.R.L.
Asuncion – Paraguay (Nov/2005-Aug/2006)

- Tasks related to electrical installations design, tanks and vessels design and piping design for several industries such as a Stevia Processing Plant, a Shipyard and an alcohol dispensing station.
- I made a very successful presentation, using 3D modeling tools, of a Stevia Processing Plant project that led to the approval of the Plant Construction.

Technical Drawer @ CORSAT S.R.L.
Asuncion – Paraguay (Jan/2005-Jul/2005)

- Calculus and 3D design of a waste dryer for meat industry; several engineering related jobs.

🦾 Mechanical Engineering Certifications

ASHRAE HVAC Design: Level I & Level II - Details
December, 2016 - ASHRAE Headquarters, Atlanta, GA, EEUU
Revit Architecture 2014 Certified Professional - Details
Autodesk University 2013 – Las Vegas Nevada

📌 On The Side

Organizer @ Energy and Enviroment Workshop (2009)
UNA University Campus, Asuncion, Paraguay.